Noise Schedule Ablation in Diffusion Models

Phase 1 Declaration, Foundation Models and Generative AI, IIT Jodhpur

Submitted by: M26AI1119 [add teammates' names and roll numbers] | Submission date: October 10, 2026

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1. Image-to-image task

Toonification / photographic-to-artistic portrait translation: converting a real photographic human face portrait into a painterly, "Old Masters" style portrait, in the stylistic domain of pre-20th-century portrait painting as curated by the MetFaces dataset, while preserving the subject's approximate facial structure and pose. This is one of the example tasks named directly in the assignment brief.

2. Pretrained model

- **Model:** google/ddpm-celebahq-256, an unconditional Denoising Diffusion Probabilistic Model (DDPM) trained on CelebA-HQ 256x256 real face photographs.
- **Paper:** Ho, Jain, Abbeel, "Denoising Diffusion Probabilistic Models", arXiv:2006.11239, NeurIPS 2020. Link: https://arxiv.org/abs/2006.11239
- **Checkpoint / codebase:** https://huggingface.co/google/ddpm-celebahq-256 (Hugging Face diffusers, `DDPMPipeline` = `UNet2DModel` + `DDPMScheduler`).

3. Dataset

- **Name:** MetFaces (NVlabs).
- **Source URL:** https://github.com/NVlabs/metfaces-dataset
- 1,336 high-quality 1024x1024 PNG images of human faces, extracted from works of art (paintings, drawings, sculptures) via the Metropolitan Museum of Art Collection API, aligned and cropped using dlib. Released under a Creative Commons BY-NC 2.0 license by NVIDIA.

4. Evidence the dataset was not used during pretraining

- **Route used:** (b), describing the pretraining data domain and arguing the chosen dataset belongs to a clearly different domain.
- **Citation:** Karras, Aila, Laine, Lehtinen, "Progressive Growing of GANs for Improved Quality, Stability, and Variation", ICLR 2018, arXiv:1710.10196, page 14, Appendix C ("The CelebA-HQ Dataset"). Figure 8 caption: "We start with a JPEG image (a) from the CelebA in-the-wild dataset." Same page, following paragraph: "We perform the above processing for all 202599 images in the dataset ... discard all but the best 30000 images."
- **Argument:** CelebA-HQ (the pretraining data for google/ddpm-celebahq-256) is built entirely from real photographic, "in-the-wild" JPEG camera photos of celebrities, filtered by image-quality metrics to the best 30,000 of 202,599 photos. MetFaces consists of painted, drawn, or sculpted portraits sourced from a museum art collection, with brushstroke, pigment, and material textures that have no camera-sensor origin, spanning historical art periods that predate photography entirely. These two datasets come from fundamentally different image-generation processes (camera optics versus hand-applied artistic media), so MetFaces lies clearly outside the CelebA-HQ pretraining domain.

5. Native noise parameterization

- theta = beta_t, the DDPMScheduler's per-step forward variance.
- Confirmed directly from the model's scheduler config (scheduler_config.json on the Hugging Face repo): `beta_schedule = "linear"`, `beta_start = 0.0001`, `beta_end = 0.02`, `num_train_timesteps = 1000`.

- So beta_t increases linearly from 0.0001 (t=1) to 0.02 (t=1000). Since 0.0001 <= beta_t <= 0.02 for all t, and 0.02 is far below 1, theta is in (0, 1) for every t. This matches the schedule described in Ho et al. 2020 for their CelebA-HQ model.

6. Chosen exponent

- **x = 2.00**
- Reasoning: since 0 < beta_t < 1 for every t, squaring shrinks beta_t further (beta_t^2 < beta_t everywhere), giving an analytically simple, easily plotted "weaker" modified schedule for Run B to contrast against Run A's original schedule in the Schedule Analysis section of the final report.

This declaration states the task, model, dataset, and schedule parameters chosen for the fine-tuning experiments (Run 0, Run A, Run B) described in the assignment brief. The full code, logs, and report are due October 25.